2**Date: 14 December, 2025**

- **Context Free Grammar**
  - An alphabet of terminal letter from which we are going to form strings of a language.
  - A set of symbols, called non-terminals
  - Finite set of productions of the form:
    - One terminal -> (produces) Strings of terminals AND / OR non-terminals
    - Non-terminals are on the left.
- **Example**
  - ANY NUMBER -> FIRST DIGIT
  - FIRST DIGIT -> FIRST DIGIT . OTHER DIGIT
  - FIRST DIGIT -> 1..9
  - OTHER DIGIT -> 0..9
- The above example cannot produce 0 (because first digit does not have 0 available)
- Single arrow (->) Symbolic representation. Double arrow (=>) when applying it numerically on an example.
- **Example*
  - S -> aS
  - S -> Λ
  - One liner: S -> aS | Λ
    - Represents: a*
- **Example**
  - S -> SS
  - S -> a
  - One liner: S -> SS | a
    - Represents: a^+
- **Example**
  - S -> aS
  - S -> bS
  - S -> a
  - S -> b
  - S -> Λ
  - One liner: S -> aS | bS | a | b | Λ
    - Represents: (a + b)*
- **Example**

    - S -> XaaX
    - X -> aX | bX | Λ
        - Represents: (a+b)*(aa)(a+b)*
- **Example**
    - S -> S.S
    - S -> Balanced.S
    - S -> S.Balanced
    - S -> Λ
    - S -> Unbalanced.S.Unbalanced
    - Unbalanced -> ab | ba
    - Balanced -> aa | bb
        - Λ, aa, bb, abab, ...
        - Represents: even-even
- **Example**
    - S -> aSa
    - S -> bSb
    - S -> a
    - S -> b
    - S -> Λ
        - One-liner: S -> aSa | bSb | a | b | Λ
        - Represents: Palindrome strings
        - S -> aSa | bSb | Λ: represents palindrome strings of even length
        - S -> aSa | bSb | a | b: represents palindrome strings of odd length
- **Example**
    - S -> aS | bb
    - Or:
        - S -> aS
        - S -> bb
    - Represents: a*bb
- **Example**
    - a*b*
    - Represented by: S -> aS | Λ, S -> Sb | Λ